 Paper No: 8

For the submission of your work:

- Create a folder named **RollNo_Name_DBI202_PaperNo**, e.g. he170145_LongNT_DBI202_01. **Do not** create any subfolder in this folder. All file created will be located in the above folder.

- For each question, you are required to write a database script. Create a file with the name corresponding to the index of the question. For example, **for question 1**, we will create a file named **Q1.sql** and create a file **Q2.sql for question 2**. So, if you do 10 questions, your folder must contain **only** 10 files Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql, Q9.sql and Q10.sql.

- Do not use any commands having the database name such as create database, alter database, use [database name], *etc.*

- Your response must contain only necessary commands for answering the question. Do not include any other command. For example, if you are required to create a trigger/procedure, then your response should contain only commands for creating the corresponding trigger/procedure; all commands for testing the created trigger/procedure are forbidden.

- Do not use GO command in your code

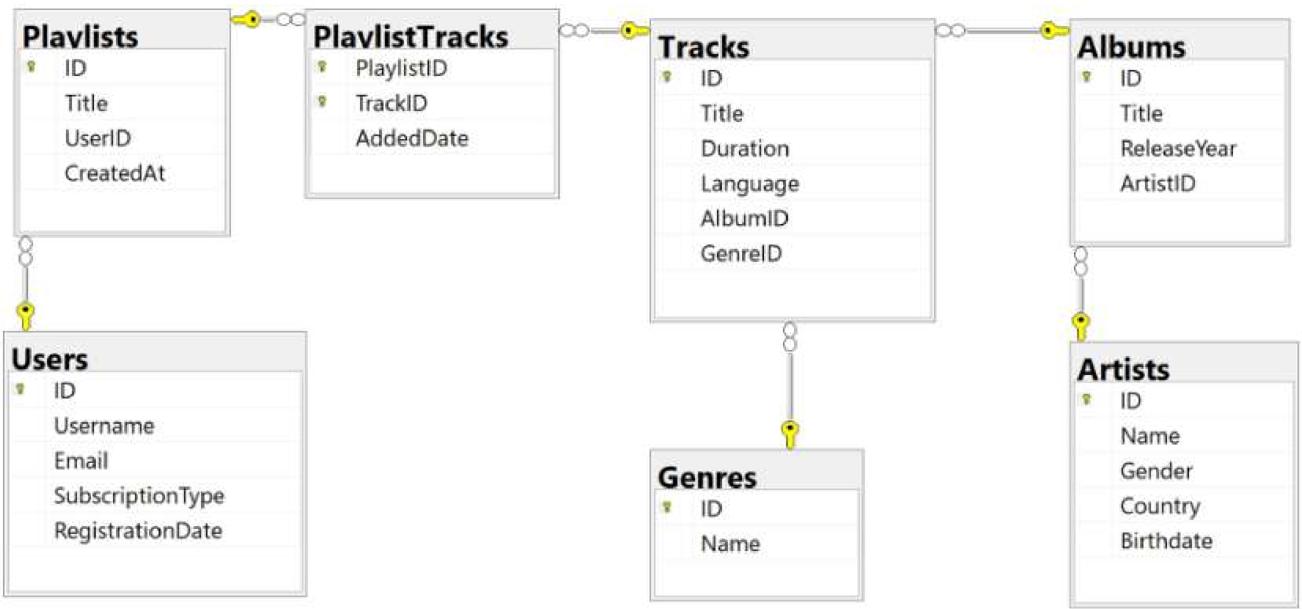
- On completion, import your work by browsing to the above folder.

- Note that:

+ You could use only SQL Server, SQL Server Management Studio, and paper or offline document in your computer.

+ If <u>any</u> of the previous requirements is not respected, your mark will be 0.

From the 2nd question, you should use the database provided in the .sql file which has the following database diagram. Please, run the provided script to create tables and insert data into your database.



Question 1:

Create one database and then write SQL statements to create, in this database, all tables derived from the ERD given in the following picture with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as request, you MUST keep the name of tables, relationship, attributes and data type of attributes as SAME as given in the given ERD.

Attributes with underline are Primary Key of each entity.

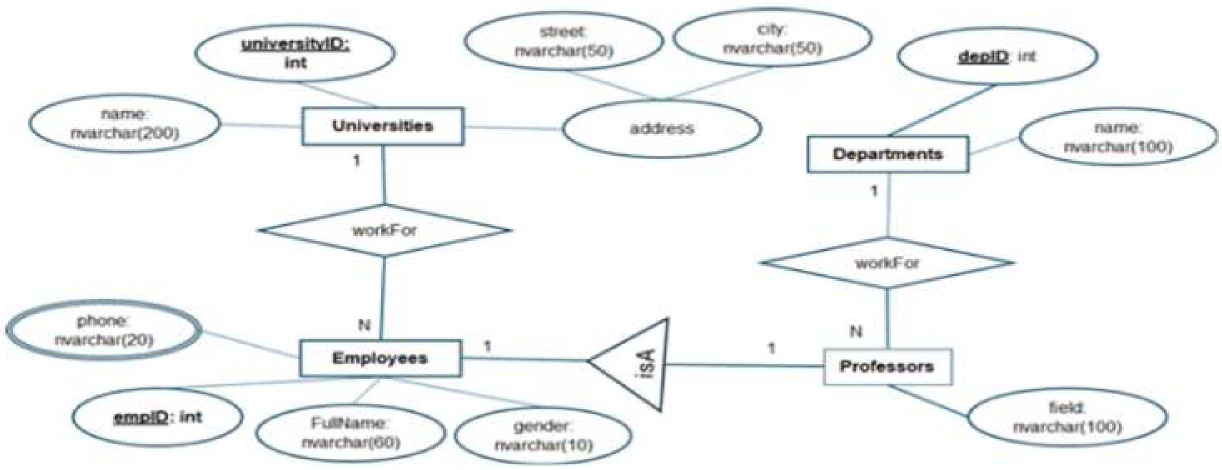
Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

If a table must be created for representing a relationship, the name of the table must be the same as the name of the relationship.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use “create database”, “Alter database”, “use database_name” or any statements using database’s name in your submission.

You must use ER style conversions for subclasses in this diagram if they exist.

The table represents the multi-values attribute must have the name which is the concatenation of the table name and the attribute name. For example, if table named Student has the multi-values attribute named Phone, the name of the corresponding table must be StudentPhone.



Picture 1.1

Question 2:

Write an SQL query to display all tracks whose title contains the string “love” (not case-sensitive), as shown in the following figure:

	ID	Title	Duration	Language	AlbumID	GenreID
1	2	Love Story	235	English	1	1
2	15	Give Me Love	285	English	5	1
3	23	Love of My Life	218	English	7	2
4	28	Make You Feel My Love	213	English	9	12
5	34	Send My Love	223	English	11	12
6	59	Fake Love	239	Korean	19	8
7	79	Lovesick Girls	192	Korean	26	8
8	141	Somebody To Love	220	English	47	1
9	145	Love Yourself	233	English	48	1
10	153	Love Poem	243	Korean	51	8
11	169	This Love	206	English	56	2
12	170	She Will Be Loved	257	English	56	2
13	178	Nobody's Love	211	English	59	2
14	191	Love On The Brain	224	English	63	4
15	199	Love The Way You Lie	263	English	66	3
16	200	No Love	299	English	66	3
17	215	Stupid Love	193	English	71	7
18	223	Love Me More	210	English	74	1
19	242	Love Galore	215	English	80	4
20	255	LOVE.	214	English	84	3
21	270	Same Old Love	229	English	89	1
22	271	Lose You To Love Me	206	English	90	1
23	302	Show Me Love	214	English	100	12

Picture 2.1

Question 3:

Write an SQL query to display all tracks that belong to albums released between 2017 and 2021 (inclusive), and which are in the genre named 'Rock'. Display the results with track's ID (rename as TrackID), track's Title (rename as TrackTitle), album's Title (rename as AlbumTitle), ReleaseYear, genre's Name (rename as GenreName) for each track, as shown below:

	TrackID	TrackTitle	AlbumTitle	ReleaseYear	GenreName
1	22	You're My Best Friend	Divide	2017	Rock
2	23	Love of My Life	Divide	2017	Rock
3	126	Next To Me	Evolve	2017	Rock
4	127	Believer	Evolve	2017	Rock
5	128	Enemy	Mercury - Act 1	2021	Rock
6	129	Wrecked	Mercury - Act 1	2021	Rock
7	130	Follow You	Mercury - Act 1	2021	Rock
8	177	Memories	Jordi	2021	Rock
9	178	Nobody's Love	Jordi	2021	Rock
10	179	Lost	Jordi	2021	Rock
11	256	Sign Of The Times	Harry Styles	2017	Rock
12	257	Kiwi	Harry Styles	2017	Rock
13	258	Sweet Creature	Harry Styles	2017	Rock
14	259	Lights Up	Fine Line	2019	Rock
15	260	Adore You	Fine Line	2019	Rock
16	261	Watermelon Sugar	Fine Line	2019	Rock

Picture 3.1

Question 4:

Write an SQL query to display all users who have a subscription type of 'Free' and registered in the year 2024, together with the playlists they created and the tracks (if any) in those playlists that belong to the genre named 'Electronic'.

Display the results with the following columns: user's ID (rename as UserID), Username, SubscriptionType, RegistrationDate, playlist's Title (rename as PlaylistTitle), track's Title (rename as TrackTitle), and genre's Name (rename as GenreName).

Include all eligible users even if they have not created any playlists and include their playlists even if they do not contain any 'Electronic' tracks.

Sort the result by Username in ascending order, then by PlaylistTitle in ascending order within the same user, and finally by TrackTitle in ascending order within the same playlist as shown below:

	UserID	Username	SubscriptionType	RegistrationDate	PlaylistTitle	TrackTitle	GenreName
1	28	ella_m	Free	2024-02-02	Evening Relax	Coloratura	Electronic
2	28	ella_m	Free	2024-02-02	Polish Vibes	Higher Power	Electronic
3	28	ella_m	Free	2024-02-02	Rock Mix	My Universe	Electronic
4	26	nora_e	Free	2024-01-05	Deep House	NULL	NULL
5	26	nora_e	Free	2024-01-05	Jordan Hits	NULL	NULL
6	26	nora_e	Free	2024-01-05	Morning Chill	Chromatica I	Electronic
7	30	zara_s	Free	2024-03-01	Canada Chill	NULL	NULL
8	30	zara_s	Free	2024-03-01	Hip Hop Canada	NULL	NULL
9	30	zara_s	Free	2024-03-01	Summer 2024	NULL	NULL

Picture 4.1

Question 5:

Write an SQL query to list all users, together with:

- The total number of playlists they created in the year 2024, and
- The total number of distinct tracks in the language 'English' in those playlists.

Display the results with: User's ID (rename as UserID), Username, the total number of playlists they created in 2024 (rename as NumberOfPlaylists), the total number of distinct English tracks in their playlists created in 2024 (rename as NumberOfTracks).

Note: The result must include all users, even if they did not create any playlists in 2024 or if their playlists created in 2024 do not contain any English tracks.

	UserID	Username	NumberOfPlaylists	NumberOfTracks
1	1	alexm	0	0
2	2	sarah_k	0	0
3	3	mike_92	0	0
4	4	emma_w	0	0
5	5	li_wei	0	0
6	6	carlos_7	0	0
7	7	amina_h	0	0
8	8	daniel_r	0	0
9	9	sofia_l	0	0
10	10	samuel_w	2	8
11	11	noah_b	0	0
12	12	olivia_j	0	0
13	13	lucas_m	0	0
14	14	fatima_z	0	0
15	15	ethan_p	0	0
16	16	mia_t	1	2
17	17	ryan_s	2	10
18	18	hannah_c	2	2
19	19	youssef_a	2	9
20	20	isabella_d	2	4
21	21	jack_l	2	6
22	22	chloe_v	2	6
23	23	ahmed_k	2	4
24	24	grace_n	3	7
25	25	leo_f	3	9
26	26	nora_e	3	9
27	27	omar_h	3	11
28	28	ella_m	3	12
29	29	adam_t	3	12
30	30	zara_s	3	11

Picture 5.1

Question 6:

Write an SQL query to display, for each artist whose Country is 'Canada', the album(s) that have the highest number of tracks among all albums of that artist.

Display the results with: artist's ID (rename as ArtistID), artist's Name (rename as ArtistName), album's ID (rename as AlbumID), album's Title (rename as AlbumTitle), total number of tracks in the album (rename as NumberOfTracks).

Note that:

- if an artist has multiple albums with the same highest number of tracks, display all of them.

- if an artist has no album, this artist will not be displayed in the results.

	ArtistID	ArtistName	AlbumID	AlbumTitle	NumberOfTracks
1	4	Drake	16	Scorpion	5
2	9	The Weeknd	28	Kiss Land	3
3	9	The Weeknd	29	Beauty Behind the Madness	3
4	9	The Weeknd	30	Starboy	3
5	9	The Weeknd	31	After Hours	3
6	13	Shawn Mendes	44	Handwritten	3
7	13	Shawn Mendes	45	Illuminate	3
8	13	Shawn Mendes	46	Wonder	3
9	14	Justin Bieber	47	My World 2.0	3
10	14	Justin Bieber	48	Purpose	3
11	14	Justin Bieber	49	Justice	3

Picture 6.1

Question 7:

Write an SQL query to display, for each country, the album(s) of artists from that country that:

- Have the highest total number of times their tracks were added to playlists (across all years) compared to all other albums in the same country,

- and also have the highest total number of times their tracks were added to playlists in the year 2024 compared to all other albums in the same country.

Display the results with the following columns:

- Country
- Album's ID (rename as AlbumID)
- Album's Title (rename as AlbumTitle)
- Total number of times the album's tracks were added to playlists across all years (rename as TotalPlaylistAdditions)

- Total number of times the album's tracks were added to playlists in 2024 (rename as TotalPlaylistAdditionsIn2024)

Order the results in ascending order of Country, and then in ascending order of AlbumTitle, as shown below.

	Country	AlbumID	AlbumTitle	TotalPlaylistAdditions	TotalPlaylistAdditionsIn2024
1	Barbados	60	Music of the Sun	3	3
2	Canada	16	Scorpion	10	7
3	South Korea	17	2 Cool 4 Skool	9	5
4	UK	10	21	8	4

Picture 7.1

Question 8:

Write SQL statement to create a table-valued function named returnTopArtistByYear that accepts one input parameter (@year int) and return a table containing, for each country, the artist(s) from that country who have the highest total number of times their tracks were added to playlists in the specified year compared to all other artists in the same country. The returned table must include Country, Artist's ID (rename as ArtistID), Artist's Name (rename as ArtistName) and the total number of times the artist's tracks were added to playlists in the specified year (rename as TotalPlaylistAdditionsInYear). If multiple artists from the same country share the highest total, all of them must be included in the result.

For example, when we execute the returnTopArtistByYear function by using the following query, the results must be shown as in the following figure:

select * from returnTopArtistByYear(2023)

	Country	ArtistID	ArtistName	TotalPlaylistAdditionsInYear
1	USA	1	Taylor Swift	12
2	UK	3	Adele	12
3	Canada	4	Drake	10
4	South Korea	5	BTS	13
5	Barbados	18	Rihanna	0

Picture 8.1

Question 9:

Create a trigger named tr_insertPlaylist on the Playlists table to validate the data before inserting rows into the table.

When an insert statement is executed on the Playlists table, the trigger will be run to enforce the following rules:

- If the UserID of the playlist to be inserted does not exist the Users table, that row must not be inserted.

- If the value of CreatedAt of the playlist to be inserted is greater than or equal to the RegistrationDate of the corresponding user and less than or equal to the current system datetime, the playlist should be inserted into the Playlists table with the provided CreatedAt value.

- If the value of CreatedAt is earlier than the user's RegistrationDate or later than the current system datetime, the playlist should still be inserted, but the CreatedAt value must be replaced with the current system datetime.

The trigger must be able to handle multiple rows inserted at the same time.

For example, if the following statement is executed to insert rows into the Playlists table:

- 'Playlist Test 1' will be inserted with the provided values because User's ID = 10 exists in the users table, and the CreatedAt value is valid.
- 'Playlist Test 2' will not be inserted because there is no user with ID = 50 in the Users table.
- 'Playlist Test 3' will be inserted with the current system datetime as the value for CreatedAt because the provided CreatedAt is earlier than the RegistrationDate of the user with the ID = 12.
- 'Playlist Test 4' will be inserted with the current system datetime as the value for CreatedAt because the provided CreatedAt is greater than the current system datetime and User's ID = 13 exists in the Users table.

```
INSERT Playlists (Title, UserID,CreatedAt)
VALUES ('Playlist Test 1',10,'2025-10-12'),
('Playlist Test 2',50,'2025-11-12'),
('Playlist Test 3',12,'2020-05-12'),
('Playlist Test 4',13,'2031-02-15')
```

<u>Question 10:</u>

Write SQL statements to insert rows into the PlaylistTracks table so that all tracks from the album titled 'Heartbreaker' are added to the playlist with ID = 11, and February 5, 2026 is used as the value for the CreatedAt, specifying only the date (do not include the time).